Jahnavi Singh

(602) 574 - 3737 | jahnavisingh6@gmail.com |linkedin

SUMMARY

Data Analyst/Data Scientist with **3.5 years** of experience in **data analytics**, **predictive modeling**, **machine learning**, **data visualization** and **cloud computing**. Proficient in **Python**, **SQL**, **Tableau**, **Power BI**, **AWS** and **Azure** to drive data-driven decision-making. Skilled in **ETL pipeline development**, **statistical analysis**, and building scalable **ML models**. Managed datasets exceeding **1M+ records** and deployed **20+ machine learning models** to improve forecasting and operational efficiency. Optimized workflows to reduce data processing time by **30%** and built **11+ interactive dashboards** that delivered actionable insights to business stakeholders.

SKILLS

Methodologies: Agile (Scrum), SDLC
Programming & Scripting: Python (NumPy, Pandas, Scikit-Learn, PyTorch, TensorFlow, spaCy, OpenCV, PyResparser), R, SQL, Bash
Data Engineering & Databases: PostgreSQL, MySQL, Snowflake, Oracle, MongoDB, Databricks, Kafka, Hadoop, Flask, REST APIs, ETL Pipelines, Apache Airflow, Data Warehousing (Snowflake, Redshift, Databricks), Data Lakes (AWS S3)
Cloud & DevOps: AWS (S3, Lambda, Redshift, EMR, Kinesis, Firehose, IAM), Microsoft Azure, Docker, Git, GitHub
Machine Learning & AI: Supervised & Unsupervised Learning, Feature Engineering, NLP (LLMs, Prompt Engineering, LangChain, OpenAI GPT APIs), Knowledge Graphs, Anomaly Detection, CNN Modeling
Statistical & Analytical Modeling: Hypothesis Testing, Regression, Time Series Forecasting, A/B Testing
Data Visualization & BI: Tableau, Power BI, Matplotlib, Seaborn, ggplot2, Advanced Excel, Data Storytelling

EXPERIENCE

Data Analyst | Blue Cross Blue Shield – USA **Aug 2024 - Present**

- Led a HIPAA-compliant, cross-functional project analyzing de-identified electronic health records (EHRs), billing systems, and patient interaction datasets to identify factors contributing to churn and disengagement among Medicare Advantage across 50 states.
- Designed and maintained ETL workflows using SSIS to extract, transform, and load large-scale clinical and administrative datasets into AWS S3, ensuring data integrity and compliance.
- Developed and fine-tuned multi-algorithm patient risk prediction models using Logistic Regression, Random Forest, and XGBoost, achieving 90% accuracy and providing actionable insights on high-risk patient segments.
- Conducted rigorous model evaluation using cross-validation, AUC-ROC curves, precision-recall analysis, and confusion matrices, continuously optimizing model performance, and increasing predictive reliability by 2%.
- Implemented data versioning and model monitoring in AWS S3 and SageMaker, tracking data lineage, detecting model drift, and enabling automated retraining to maintain predictive accuracy over time for clinical outcomes.
- Built comprehensive Power BI dashboards visualizing patient risk trends, disease progression, and intervention effectiveness, effectively translating complex analyses for 25+ healthcare stakeholders and influencing strategic care decisions.

Data Analyst | Accenture – India **Apr 2021 – Jan 2023**

- Analyzed historical sales and inventory data of 100,000+ SKUs to identify demand patterns, seasonality, and key factors influencing stock levels, leading to a 15–20% improvement in inventory planning decisions.
- Executed ETL pipelines using Python (Pandas, NumPy) and SQL, automating data preprocessing, cleaning, and feature engineering (e.g., demand trends, stock-out frequency, lead time) for over 20+ variables, reducing manual effort by 80%.
- Conducted variable transformations, correlation analysis, and statistical tests (ANOVA, Chi-Square, U-Test, T-Test) to uncover relationships between sales, promotions, seasonal trends, and supply constraints.
- Developed Machine Learning models (Linear Regression, Random Forest, Gradient Boosting) to forecast SKU-level demand, achieving predictive accuracy of 86.3%.
- Optimized model performance using Grid Search, Random Search, and PCA, improving overall forecast accuracy by 8%.
- Performed A/B testing on promotional strategies and inventory interventions to evaluate their impact on sales and stock efficiency, providing evidence-based recommendations for process improvements.
- Deployed forecasting models via FastAPI/Flask, enabling batch-wise inventory predictions for 20k+ SKUs per quarter.
- Built interactive Tableau dashboards, visualizing forecasted demand, stock levels, and reorder alerts for 25+ stakeholders, driving actionable insights for procurement and supply chain planning.

PROJECTS

AI-Powered Resume Optimizer with LLM

- Built a full-stack AI-driven resume optimization tool using Python, spaCy, TF-IDF, Flask, AWS, and PostgreSQL; parsed resumes and job descriptions with NLP to compute match scores and skill gaps.
- Leveraged OpenAI GPT APIs and LangChain for LLM-based content generation and prompt-engineered personalized improvement suggestions.
- Deployed scalable cloud architecture on AWS with PostgreSQL and Snowflake; achieved ~85% skill extraction accuracy with sub-3 second match updates.

School Similarity Matching & Clustering Model

- Developed a similarity-based model (KNN, cosine similarity) achieving 89% accuracy to classify schools by performance metrics, followed by K-Means clustering and t-SNE for anomaly grouping and benchmarking insights.

EDUCATION

- **M.S., Information Technology (Information Systems Management)** – Arizona State University, Tempe, AZ | May 2025
  GPA: 3.87/4.0
- **B.Tech., Electronics and Telecommunications** – Symbiosis Institute of Technology, Pune, India | May 2023